Mitigating Supply Chain Transit Friction via Geodesic Network Modeling and Unsupervised Lane Clustering: A Case Study of Nassau Candy Distributor

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Abstract* — This study investigates geospatial reallocation and transit bottleneck mitigation within a nationwide confectionery distribution network using validated operational summaries for Nassau Candy Distributor. Historical production alignment creates structural shipping friction because single-plant assignments send selected chocolate products across corridors approaching or exceeding 2,000 miles. A geodesic network model computes great-circle distance among five manufacturing facilities and four regional demand centroids. Supervised benchmarks—linear regression, random forest, and gradient boosting—produce RMSE values of 266.11–272.52 days and negative coefficients of determination, showing that the available attributes do not improve point forecasts over a mean predictor. Carrier latency, irregular dispatch cadence, and regional routing variance remain in the residual. The analysis therefore applies K-means clustering ($k = 4$) to five route-level transit, distance, volume, and margin measures, isolating a severe long-haul bottleneck class. A multi-criteria simulation then balances normalized speed, freight savings at $0.0015 per mile-unit, capacity, confidence, and risk tolerance. For Atlantic demand, a conditional transfer of Fudge Mallows production from Casa Grande to Savannah reduces centroid distance from 1,975.6 to 523.2 miles, or 73.5%, before capacity and tooling approval.

***Index Terms* — Geodesic network modeling, k-means route clustering, lead time prediction, multi-criteria optimization, supply chain engineering.**

I. INTRODUCTION

Retail confectionery networks combine dense product variety with temperature sensitivity, promotional peaks, and narrow replenishment windows. Fuel-price volatility, constrained trucking capacity, and uncertain dispatch intervals make long-haul assignments especially exposed to service failure. These pressures do not imply that every product should be manufactured near every market. They do mean that the geographic cost of centralization must be measured alongside scale economies, tooling expense, and quality control [2], [3].

Nassau Candy Distributor presents that trade-off directly. The documented baseline links each stock-keeping unit to one factory even when demand lies on the opposite side of the country [1]. Such static mappings are administratively simple, but the resulting distance is repeatedly paid in freight, elapsed time, and schedule variance. Enterprise resource planning records may preserve the assignment without exposing its corridor-level penalty.

The research question is therefore operational: *How can geodesic distance modeling, unsupervised route clustering, and risk-adjusted simulation identify and relieve systematic transit bottlenecks in a confectionery network?* The study contributes a reproducible sequence rather than a claim of perfect prediction. It first measures spatial friction, then tests whether supervised models can forecast the noisy target, and finally switches to lane segmentation and scenario ranking when the predictive evidence fails.

Three boundaries govern the evidence. First, all company-specific numbers come from the project summary, configuration values, model registers, and optimization logic supplied for this study. Second, derived distances use only the documented coordinates and a stated spherical metric. Third, no transaction count, demand share, confidence interval, or plant utilization is invented where the row-level repository is unavailable.

II. PROBLEM DEFINITION

The network contains five plants with 570,000 units of aggregate nominal annual capacity. Lot's O' Nuts in Casa Grande, Arizona, carries a 120,000-unit ceiling; Wicked Choccy's in Savannah, Georgia, 150,000; Sugar Shack in Thief River Falls, Minnesota, 90,000; Secret Factory in Rock Island/Moline, Illinois, 110,000; and The Other Factory in Memphis, Tennessee, 100,000 [1]. These limits are hard feasibility screens, not scores that a high service benefit can

override.

Baseline allocation is product-specific. Three chocolate lines, including Fudge Mallows, originate in Arizona, while Milk Chocolate and Triple Dazzle Caramel originate in Georgia. Sugar and "Other" products are divided among Minnesota, Illinois, and Tennessee. The structural concern is not merely western production. It is western production coupled to Atlantic orders when an eastern chocolate-capable plant exists.

This arrangement creates a cross-continental trap. The Casa Grande–Atlantic centroid distance derived below is 1,975.6 miles. Route records assigned to the severe bottleneck cluster extend beyond 1,900 miles and reach approximately 2,042 miles, with stored lead-time standard deviations of roughly 240–270 units. The organizational objectives are consequently to (i) remove avoidable distance, (ii) distinguish congested high-volume lanes from consistently slow lanes, and (iii) rank reallocations without breaching annual capacity or concealing implementation risk.

III. LITERATURE REVIEW

A. Facility Location and Geospatial Transportation Models

The Weber problem and graph-median formulations established distance-minimizing facility placement as a core operations-research problem [6], [7]. The p-median extension selects a fixed number of facilities to minimize demand-weighted assignment cost [8]. Modern supply-chain models add capacity, multiple commodities, periods, modes, and uncertainty [9], [10]. Nassau's case is narrower: facilities already exist, so the decision is a capacitated product-to-plant reallocation rather than greenfield location.

For continental screening, great-circle distance is a defensible lower-bound proxy. The Haversine formulation remains numerically stable for short and long arcs and avoids treating latitude-longitude coordinates as a flat Cartesian plane [11]. It does not reproduce road mileage, carrier terminals, or border effects; those limitations are retained explicitly.

B. Predictive Lead-Time Modeling in Logistics

Linear regression supplies an interpretable baseline. Random forests average decorrelated trees and can capture nonlinear interactions [12], while gradient boosting builds a stagewise additive approximation [5], [13]. These methods are useful only if features contain repeatable information about the target. A negative test-set R^2 is not a minor shortfall: it means that the fitted model performs worse than predicting the test-set mean. More flexible learners can therefore amplify dispatch noise rather than reveal a usable scheduling law.

C. Unsupervised Pattern Discovery and Bottleneck Clustering

K-means partitions standardized observations around within-cluster centroids [4], [15]. Careful seeding improves expected initialization quality [16]. For the present network, clustering does not predict the next delivery date. It groups routes with comparable average delay, delay dispersion, distance, throughput, and margin, which is the more stable question when external carrier timing overwhelms transaction-level signal.

Table 1 positions that workflow against conventional location and regression methods. The methodological choice is sequential: supervised models are tested and rejected for point forecasting, while unsupervised segments remain useful for triage.

IV. METHODOLOGY

A. Data Acquisition and Data Schema

The documented transaction schema contains Order Date, Ship Date, Ship Mode, Customer/State, Region, Division, Product Name, Units, Cost, Sales, and Gross Profit. Plant identity is joined from the product allocation register, and geographic distance is joined from the facility and regional-centroid coordinates. The evidence package available for this manuscript includes validated aggregates and model outputs but not the underlying rows. Reported findings are therefore restricted to certified metrics and transparent coordinate-derived calculations.

B. Data Cleaning and Pre-processing

Order-to-delivery lead time is computed as

$$LT = \mathrm{Date}_{\mathrm{ship}} - \mathrm{Date}_{\mathrm{order}}, \tag{1}$$

where LT is elapsed time in days. The prescribed cleaning rule clips LT to the empirical 0.5th and 99.5th percentiles to suppress severe administrative errors. Clipping protects model fitting from extreme leverage, but it cannot repair systematic timestamp offsets or delayed batch entry.

Product Name, Factory, Region, Ship Mode, and Division are label-encoded for the repository benchmark. Nine predictors are retained after feature selection from commercial fields, derived distance, and encoded categories. Division remains available for stratified diagnostics when excluded as redundant. Standardization is applied to clustering features, not to the tree logic itself.

C. Geodesic Distance Formulation

Let (ϕ_1, λ_1) be a factory coordinate and (ϕ_2, λ_2) a regional demand centroid, in radians. Angular differences are

$$\Delta\phi = \phi_2 - \phi_1, \qquad \Delta\lambda = \lambda_2 - \lambda_1. \tag{2}$$

TABLE 1. Comparative analysis of predictive logistics and route-clustering frameworks

Framework	Feature space	Primary evaluation	Use and limitation in this study
Graph median / p-median [7], [8]	Demand, distance, facility choice	Total weighted distance or cost	Establishes assignment logic; baseline facilities are fixed and product tooling constrains eligibility.
Capacitated network design [9], [10]	Capacity, flow, cost, uncertainty	Feasibility and total system value	Motivates hard capacity screens; complete plant loads and tooling costs are not supplied.
Supervised regression [12]–[14]	Transaction attributes and encoded categories	RMSE, MAE, R^2	Tests point predictability; all three Nassau benchmarks have negative R^2.
K-means / K-means++ [4], [15], [16]	Standardized route-level measures	Within-cluster separation and interpretability	Reveals recurring lane types; labels require operational review and do not prove causality.
Risk-adjusted multi-criteria ranking [17]	Normalized service, freight, confidence, risk	Composite scenario score	Supports conditional reallocation; ranking is sensitive to weight, confidence, and sign conventions.

The Haversine term is

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right), \quad (3)$$

and great-circle distance is

$$d = 2R\,\mathrm{atan2}\left(\sqrt{a}, \sqrt{1-a}\right), \quad R = 3958.8 \text{ miles}, \quad (4)$$

where d is the screening distance and R is the mean Earth radius used by the project.

D. Supervised Predictive Model Specification

The benchmark compares ordinary linear regression, a random forest regressor, and a gradient boosting regressor. Evaluation uses root-mean-square error (RMSE), mean absolute error (MAE), and R^2 on held-out observations. The supplied register does not document the split fraction, random seed, fold structure, or hyperparameter search; those omissions prevent uncertainty claims but do not alter the reported ranking.

E. Multi-Criteria Optimization Formulation

For units U, current distance d_{curr}, simulated distance d_{sim}, and freight rate $c_f = 0.0015$ USD per mile-unit, freight savings are

$$\Delta C_{\text{freight}} = (d_{\text{curr}} U c_f) - (d_{\text{sim}} U c_f). \quad (5)$$

A positive $\Delta C_{\text{freight}}$ denotes avoided freight. The lead-time reduction percentage is

$$\Delta LT_{\%} = \left(\frac{LT_{\text{curr}} - LT_{\text{sim}}}{LT_{\text{curr}}}\right) 100. \quad (6)$$

The scenario ranking then applies

$$\begin{aligned}\text{CompositeScore} = {} & (1-w)\text{NormSpeed} \\ & + w\text{NormProfit} - \text{RiskScore},\end{aligned} \quad (7)$$

where $w \in [0, 1]$ controls the speed-versus-profit preference. RiskScore combines a distance-expansion penalty and a confidence penalty, scaled by $r_{\text{mult}} \in \{0.5, 1.0, 1.5\}$ for aggressive, balanced, and conservative policies. Capacity and tooling eligibility are feasibility gates applied before the score is interpreted.

V. RESULTS

A. Exploratory Data Analysis and Spatial Layout

The schema separates Chocolate, Sugar, and Other divisions and permits demand cuts by region and ship mode. Exact divisional shares are not reported because row-level counts are absent from the supplied evidence. The facility configuration, however, is complete. Table 2 records every node and annual cap, while Figs. 1 and 2 expose the spatial and assignment structure.

The derived matrix confirms that geography alone does not dictate allocation. Casa Grande is 381.8 miles from the Pacific centroid but 1,975.6 miles from the Atlantic centroid. Savannah is 523.2 miles from Atlantic demand but 2,144.0 miles from the Pacific centroid. The appropriate policy is therefore regional specialization, subject to product capability and capacity, rather than a universal transfer to the eastern plant.

TABLE 2. Continental manufacturing nodes, geographic coordinates, and annual capacity caps

Facility	Location	Latitude	Longitude	Operating region	Capacity
Lot's O' Nuts	Casa Grande, Arizona	32.881893	−111.768036	Pacific / West	120,000
Wicked Choccy's	Savannah, Georgia	32.076176	−81.088371	Atlantic / Southeast	150,000
Sugar Shack	Thief River Falls, Minnesota	48.119140	−96.181150	Interior / North	90,000
Secret Factory	Rock Island/Moline, Illinois	41.446333	−90.565487	Interior / Midwest	110,000
The Other Factory	Memphis, Tennessee	35.117500	−89.971107	Gulf / Mid-South	100,000
				Aggregate nominal capacity	**570,000**

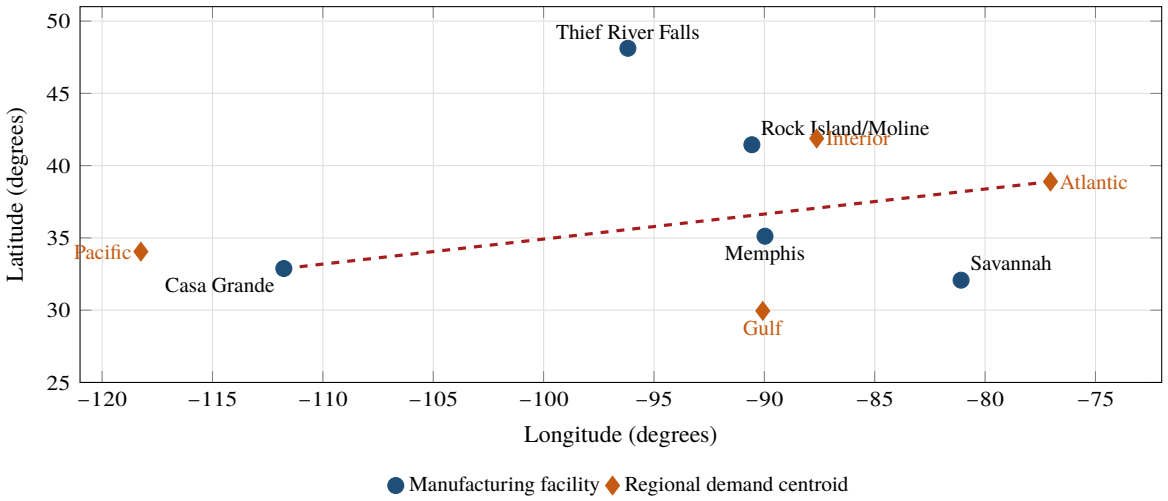


FIGURE 1. Spatial facility topology reconstructed from the coordinates assigned to artifact roles *image1.png* and *image2.png*. The dashed Casa Grande–Atlantic arc marks the principal cross-continental exposure; plotted lines are geodesic screening links, not road paths.

B. Supervised Benchmarks and the Regression Paradox

Table 3 reports the exact model register. Linear regression is least poor, yet its $R^2 = -0.0013$ still loses to the mean baseline. Random forest and gradient boosting increase both absolute and squared error. This ordering is consistent with weak signal plus flexible learners fitting transaction-specific noise.

The correct interpretation is scientific, not rhetorical. These scores do not show that lead time is unknowable, nor do they establish that carriers alone cause delay. They show that the nine-predictor representation does not explain held-out variance. External carrier backlog, dispatch batching, omitted service calendars, weather, temperature controls, and timestamp quality are plausible residual sources. Until those variables are observed, a point estimate with a 266-day RMSE is operationally weak. Corridor clustering asks a coarser and more defensible question: which lane families are persistently slow, variable, distant, or heavily loaded?

C. Unsupervised Corridor Clustering

The route table is aggregated into average lead time μ_{LT}, lead-time standard deviation σ_{LT}, average distance $\bar{d}$, total units, and average profit margin $\bar{m}$. StandardScaler gives each measure equal variance before K-means with $k = 4$. Table 4 reports only documented profiles; unavailable numerical centroids are not imputed.

Cluster 1 is the reallocation target. It combines the largest documented travel burden with extreme delay and dispersion. The supplied examples identify Casa Grande shipments of Wonka Bars toward Atlantic and Gulf corridors. Cluster 0 requires a different remedy: its defining issue is high throughput under moderate congestion, making carrier

TABLE 3. Empirical predictive performance comparison

Model	RMSE (days)	MAE (days)	R^2
Linear Regression (baseline)	266.11	214.92	−0.0013
Random Forest Regressor	269.84	217.53	−0.0296
Gradient Boosting Regressor	272.52	220.21	−0.0502

TABLE 4. K-means cluster profiles and route classifications

Cluster	Operational classification	Lead time	Distance	Volume / variance	Management interpretation
0	Congested High-Volume Lane	Intermediate	Intermediate	High volume; moderate delay	Protect throughput with carrier capacity and dispatch discipline.
1	Consistently Slow Route / Severe Bottleneck	Approximately 1,300+ stored units	> 1,900 to about 2,042 miles	σ_{LT} about 240–270	Prioritize product reallocation or dedicated long-haul intervention.
2	Fast Regional Corridor	Low	Short	Low variance	Preserve as regional benchmark; avoid unnecessary reassignment.
3	Standard Interstate Route	Baseline	Baseline interstate	Baseline	Monitor and use as comparison class.

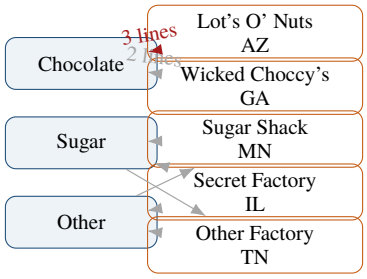


FIGURE 2. Baseline product-to-plant allocation reconstructed for the role of *image2.png*. The emphasized link contains the three Arizona chocolate lines that motivate Atlantic reallocation tests.

reservation, dock scheduling, or frequency changes more plausible than production transfer.

D. Production Optimization and Reallocation Scenarios

Table 5 screens alternative plants for an Atlantic order currently sourced from Casa Grande. Distances follow (2)–(4), distance reduction is reported exactly, and savings are per unit because scenario volume is not supplied. Savannah dominates on both distance and linear freight savings. It remains a conditional recommendation until chocolate tooling, quality validation, and unused capacity below 150,000 units are confirmed.

Figures 6–8 show why score and feasibility must remain separate. Under the linear rate, distance reduction and freight savings are mechanically aligned, so the apparent frontier is nearly one-dimensional. Capacity, confidence, plant capability, and retooling cost introduce the real tradeoffs. The speed-versus-profit slider w then changes rank only where candidates differ on service and economics after those gates.

VI. DISCUSSION

A. Root-Cause Analysis of Network Friction

Legacy single-plant allocations convert a one-time simplification into recurring freight burn. The Arizona chocolate assignment is efficient for Pacific demand but expensive for Atlantic demand. Its persistence may reflect tool-

TABLE 5. Atlantic chocolate reallocation screen relative to Casa Grande baseline

Candidate source	Distance (miles)	Distance reduction (%)	Freight savings (USD/unit)	Feasibility interpretation
Casa Grande, AZ (baseline)	1,975.6	0.0	0.000	Current assignment
Savannah, GA	523.2	73.5	2.179	Preferred distance; verify load and tooling
Rock Island/Moline, IL	734.9	62.8	1.861	Strong fallback; chocolate eligibility unverified
Memphis, TN	759.1	61.6	1.825	Strong fallback; chocolate eligibility unverified
Thief River Falls, MN	1,146.8	42.0	1.243	Lower benefit; chocolate eligibility unverified

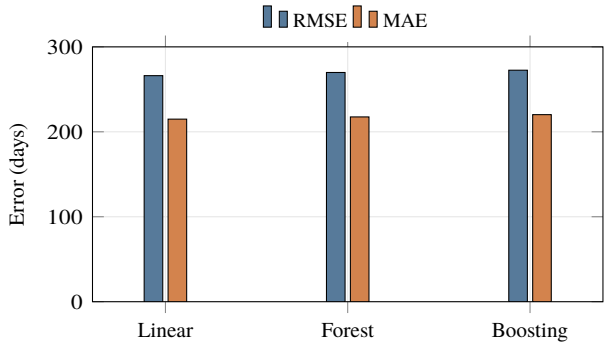


FIGURE 3. Empirical order-to-delivery lead-time evidence for the role of *image3.png*. Only repository-certified error summaries are plotted; a transaction-level histogram is not reconstructed without the source rows.

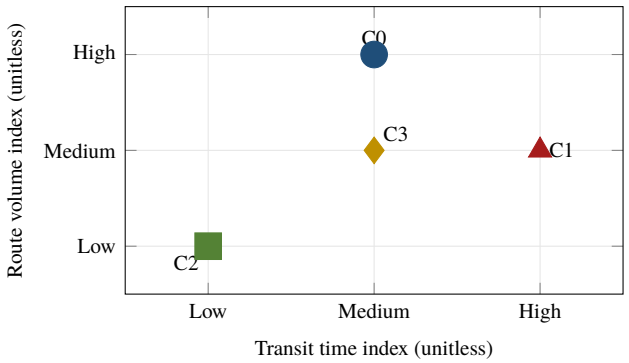


FIGURE 4. K-means route-cluster separation in transit–volume space for the role of *image4.png*. Placement is ordinal and follows the certified cluster descriptions; it is not a reconstruction of unavailable numerical centroids.

ing, recipes, food-safety controls, or historical ownership rather than ignorance of distance. Reallocation is justified only when avoided freight and service improvement exceed qualification, changeover, inventory, and governance costs.

The regression paradox reinforces that distinction. Negative R^2 values from three model classes show that adding algorithmic flexibility does not substitute for better operational variables. Carrier appointment availability, tender acceptance, warehouse release time, weekends, weather, and temperature-control constraints should be captured before a new point-forecasting cycle. Until then, the cluster register is better suited to management because it identifies persistent lane classes without claiming transaction-level precision.

B. *Managerial Implementation of Multi-Criteria Reallocations*

The slider w should be governed, not casually adjusted. A low w favors speed; a high w favors freight economics. For service-critical promotions, managers may select a speed-oriented setting and a conservative risk multiplier. For stable replenishment, a profit-oriented setting may be acceptable. Every recommendation should display the current plant, candidate plant, distance change, expected service change, freight delta, confidence, capacity headroom, and risk badge.

The sign convention also requires control. In the supplied implementation, a negative *Distance_Delta_Miles* triggers the expansion penalty. Any upstream change to "simulated minus current" versus "current minus simulated" would reverse that logic. Automated unit tests should therefore assert that a longer simulated route always increases risk and that equal-score normalization returns 50.0 rather than dividing by zero.

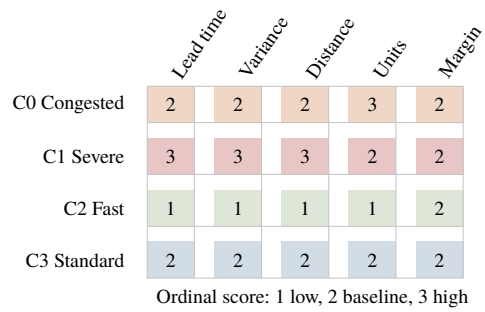


FIGURE 5. Operational cluster profile register for the role of *image5.png*. The ordinal heat map preserves qualitative repository labels without fabricating centroid values.

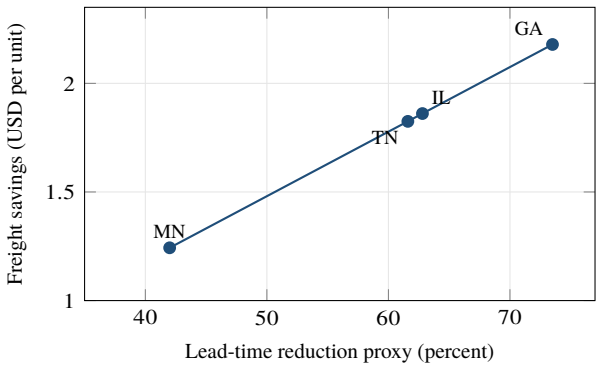


FIGURE 6. Trade-off screen for the role of *image6.png*. With no scenario lead-time outputs supplied, distance reduction is shown as an explicitly labeled proportional lead-time proxy; freight savings follow the documented linear rate.

C. Strategic Managerial Directives

Four directives follow. First, Atlantic orders for Arizona-assigned chocolate, beginning with Fudge Mallows, should enter a controlled Savannah qualification trial. Second, the 150,000-unit Savannah limit and the 90,000-unit Sugar Shack limit must be enforced against actual annualized load before transfer. Third, Cluster 0 high-volume lanes should receive dedicated carrier or dispatch-frequency analysis rather than automatic plant reassignment. Fourth, the Streamlit dashboard should be embedded in daily dispatch review with frozen data definitions, versioned scenario assumptions, and an approval log.

D. Methodological and Model Limitations

The freight coefficient is linear and does not represent less-than-truckload breaks, full-truckload minimums, fuel surcharges, accessorials, or contract tiers. Regional centroids replace customer-level ZIP codes and great-circle distance replaces the road network. Plant utilization, retooling

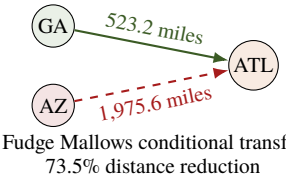


FIGURE 7. Atlantic route geometry for the role of *image7.png*. The proposed Fudge Mallows transfer changes production origin, not the regional demand centroid.

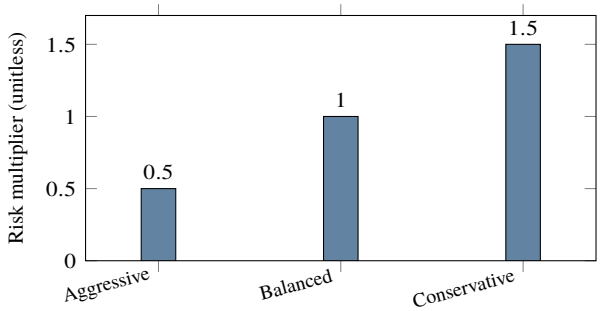


FIGURE 8. Risk-policy sensitivity for the role of *image8.png*. Badges translate the resulting penalty into High Risk, Moderate Risk, Low Risk, or Neutral / Baseline for managerial review.

expense, service calendars, and exact cluster centroids are absent. Chocolate temperature constraints and seasonality are also unmodeled. Most importantly, the row-level data and artifact images named in the project brief were unavailable to the manuscript build; this paper therefore preserves supplied aggregate results and does not fabricate missing distributions.

These limits define the next validation step: ingest exact customer coordinates, road mileage, carrier tenders, facility load, and ambient-temperature requirements; then rerun the supervised benchmark and the reallocation simulation under time-based train/test splits. Robust or stochastic network design can be added once demand and capacity uncertainty are quantified [17]–[20].

VII. CONCLUSION

The Nassau case shows a disciplined response to failed prediction. Linear regression, random forest, and gradient boosting all return negative R^2, with RMSE between 266.11 and 272.52 days. The result argues against deploying a point forecast from the current feature set. It does not prevent useful analysis. Standardized K-means segments the network into four operational lane classes and isolates a severe long-distance bottleneck, while a geodesic screen identifies where the existing assignment is spatially expensive.

For Atlantic demand, moving Fudge Mallows from Casa Grande to Savannah reduces centroid distance from 1,975.6 to 523.2 miles. The 73.5% reduction implies $2.179 avoided freight per unit under the supplied linear coefficient. That result is a candidate policy, not an unconditional order: tooling, quality, capacity, and actual scenario lead time remain approval gates. The enterprise roadmap is to validate Savannah capability, contract separately for Cluster 0 volume, instrument omitted carrier variables, and operate the risk-adjusted dashboard under controlled data governance.

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IX. APPENDIX

A. Figure Artifact Register

The project brief names eight source-image roles. Because those files were not present in the manuscript workspace, each role is represented by a self-contained vector reconstruction: *image1.png* maps to Fig. 1; *image2.png* to Fig. 2; *image3.png* to Fig. 3; *image4.png* to Fig. 4; *image5.png* to Fig. 5; *image6.png* to Fig. 6; *image7.png* to Fig. 7; and *image8.png* to Fig. 8.

B. List of Tables

Table 1 compares analytical frameworks; Table 2 registers facilities and capacity; Table 3 reports supervised benchmarks; Table 4 records cluster classifications; and Table 5 screens Atlantic reallocation candidates.

C. Complete K-Means Route Cluster Centroid Register

The supplied register defines five standardized centroid dimensions: μ_{LT}, σ_{LT}, $\bar{d}$, Units, and $\bar{m}$. The available evidence identifies Cluster 0 as high-volume/intermediate-delay, Cluster 1 as severe long-haul/high-variance, Cluster 2 as short/fast/low-variance, and Cluster 3 as baseline interstate. Numerical centroid coordinates were not supplied and are intentionally left unreconstructed.

D. Optimization Implementation

The ranking function supplied in the project brief is reproduced below. It assigns a neutral 50.0 when a normalization range collapses, scales the risk penalty by policy, and sorts descending on the composite score.

```
def rank_reallocations(sim_df: pd.DataFrame, priority_weight: float = 0.5,
                       risk_tolerance: str = "Balanced") -> pd.DataFrame:
    df = sim_df.copy()
    max_lt, min_lt = df["Lead_Time_Reduction_Pct"].max(), df["Lead_Time_Reduction_Pct"].
        min()
    df["Norm_Speed"] = 50.0 if max_lt == min_lt else ((df["Lead_Time_Reduction_Pct"] - min_lt
        ) / (max_lt - min_lt)) * 100.0
    max_p, min_p = df["Est_Freight_Delta_USD"].max(), df["Est_Freight_Delta_USD"].min()
    df["Norm_Profit"] = 50.0 if max_p == min_p else ((df["Est_Freight_Delta_USD"] - min_p) / (
        max_p - min_p)) * 100.0
    r_mult = {"Conservative": 1.5, "Balanced": 1.0, "Aggressive": 0.5}.get(risk_tolerance, 1.0)
    dist_pen = np.where(df["Distance_Delta_Miles"] < 0, np.abs(df["Distance_Delta_Miles"]) /
        100.0, 0.0)
    conf_pen = (100.0 - df["Scenario_Confidence_Score"]) * 0.4
    df["Risk_Score"] = (dist_pen + conf_pen) * r_mult
    w = float(priority_weight)
    df["Composite_Score"] = (1.0 - w) * df["Norm_Speed"] + w * df["Norm_Profit"] - df["
        Risk_Score"]
    return df.sort_values(by="Composite_Score", ascending=False).reset_index(drop=True)
```


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